

ST2334 Cheat Sheet, AY26/27 S1

Zhao Wu

September 27, 2026

1 Basic Concepts of Probability

1.1 Experiments, sample spaces, and events

- **Statistical Experiment** is a procedure that produces an observation.
 - **Sample Space** is the set of all possible outcomes of a statistical experiment.
 - **Event** is a subset of the sample space.
 - **Sample Point** is a single outcome of a statistical experiment.
- Sample Point $\subseteq$ Event $\subseteq$ Sample Space

Sure Event and Null Event

- **Sure Event** = Sample Space, S
- **Null Event** = Empty Set / Complement of Sure Event

1.2 Event Operations & Relationships

- **Union** of A and B, $(A \cup B)$
 $A \cup B = \{x \in U \mid x \in A \vee x \in B\}$
- **Intersection** of A and B, $(A \cap B)$
 $A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$
- **Difference** of B minus A, $(B - A)$
 $B - A = \{x \in U \mid x \in B \wedge x \notin A\} = B \cap \bar{A}$
- **Complement** of A ($\bar{A}$)
 $\bar{A} = \{x \in S \mid x \notin A\}$

- **Mutually Exclusive or Disjoint** Events
 $A \cap B = \emptyset$

Contained and Equivalent

- **Contained C**
 $A \subset B \Leftrightarrow \forall x, x \in A \Rightarrow x \in B$

- **Equivalent =**
 $A = B \Leftrightarrow A \subset B \wedge B \subset A$

Subset Relations (Selected)

No.	Law	Identities
7	De Morgan's laws	$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$
8	Absorption laws	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

1.3 Counting Methods

Multiplication and Addition Principles

- **Multiplication Principle:** Sequence of r stages
 $n_1 \times n_2 \times \cdots \times n_r = n_1 n_2 \cdots n_r$
- **Addition Principle:** k non-overlapping procedures
 $n_1 + n_2 + \cdots + n_k = n_1 + n_2 + \cdots + n_k$

Factorial, Permutations, and Combinations

- **Factorial** ($n!$)

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1$$

- **Permutation** (P_r^n): Selection and arrangement of r objects from n objects

$$P_r^n = \frac{n!}{(n-r)!}$$

- **Combination** ($\binom{n}{r}$): Selection of r objects from n objects

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

1.4 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative frequency of A, when experiment is repeated n times and event A occurs m times, approaches $P(A)$ as $n \rightarrow \infty$:

$$P(A) = \lim_{\{n \rightarrow \infty\}} \frac{m}{n}$$

$$0 \leq P(A) \leq 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Properties of Probability

Prop.	Name	Statement
1.4	Empty set	$P(\emptyset) = 0$
1.5	Finite additivity	If $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(A_1 \cup \dots \cup A_n) = P(A_1) + \dots + P(A_n)$
1.6	Complement	$P(\bar{A}) = 1 - P(A)$
1.7	Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$
1.8	Inclusion-exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
1.9	Monotonicity	$A \subset B \Rightarrow P(A) \leq P(B)$
1.10	Union bound (Boole's inequality)	$P(A_1 \cup \dots \cup A_n) \leq P(A_1) + \dots + P(A_n)$

1.5 Conditional Probability

Conditional Probability ($P(A \mid B)$) is probability of A occurring given that B occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B \mid A)}{P(B)}$$

Simpson's Paradox

Phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. Or formally,

$$P(A \mid B \cap C_i) \geq P(A \mid B' \cap C_i) \forall i \text{ but } P(A \mid B) < P(A \mid B')$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)P(B) \text{ if } P(B) > 0$$

1.6 Independence

Events A and B are **independent**, $A \perp B \Leftrightarrow$

$$P(A \cap B) = P(A)P(B)$$

Tautologies:

- Suppose $P(A) > 0$ and $P(B) > 0$. If $A \perp B$, then A and B are not mutually exclusive (aka $A \cap B \neq \emptyset$)
- $S \perp A$ for all events A; $\emptyset \perp A$ for all events A
- If $A \perp B$, then $A \perp B'$, $A' \perp B$, $A' \perp B'$
- Independence cannot be expressed in terms of venn diagram

1.7 Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{i=1}^n P(B \mid A_i)P(A_i)}$$

Law of Total Probability

Suppose $A_1, A_2, \dots, A_n$ is a partition of the sample space S

$$P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B \mid A_i)P(A_i)$$

$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

2 Random Variable

Random Variable is a function that transform a sample from sample space to a real number.

$$X: S \rightarrow \mathbb{R}: s \in S \text{ and } X(s) = x \in \mathbb{R}$$

Range space of X is set of real numbers that satisfies

$$R_X = \{x \mid X(s) = x \forall s \in S\}: R_X \in \mathbb{R}$$

Subsets of Sample Space

$$\{X = x\} = \{s \in S: X(s) = x\}, \{X = x\} \in S$$

$$\{X \in A\} = \{s \in S: X(s) \in A\}, \{X \in A\} \in S$$

2.1 Probability Distribution

Probability Mass Function (pmf)

Discrete random variable X_D , when $R_X = \{x_1, x_2, \dots\}$ is finite or countable.

pmf, f(x) for $\{X = x\}$

$$f(x) = \begin{cases} P(X = x) & \forall x \in R_X \\ 0 & \forall x \notin R_X \end{cases}$$

Well-defined pmf for discrete random variable X:

1. $\forall x_i \in R_X, f(x_i) \geq 0$
2. $\forall x_i \notin R_X, f(x_i) = 0$
3. $\sum_{i=1}^{\infty} f(x_i) = \sum_{x_i \in R_X} f(x_i) = 1$

Probability Density Function (pdf)

Continuous random variable X_C , when $R_X \in [a, b]$ is an interval(s).

pdf, $P(a \leq X \leq b)$ for any a and b such that $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Well-defined pdf for continuous random variable X:

1. $\forall x \in R_X, f(x) \geq 0$
2. $\forall x \notin R_X, f(x) = 0$
3. $\int_{R_X} f(x)dx = 1$

Other key properties,

- Specific point is zero: $P(X = x_i) = \int_{x_i}^{x_i} f(x)dx = 0$
- Endpoint doesn't matter: $P(a \leq X \leq b) = P(a < X < b)$

2.2 Cumulative Distribution Function

Cumulative Distribution Function (cdf), F(x) for any random variable X is defined as

$$F(x) = P(X \leq x) = \begin{cases} f(0) + f(1) + \dots + f(x) & \text{if X is discrete} \\ \int_{-\infty}^x f(t)dt & \text{if X is continuous} \end{cases}$$

$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a -)$$
$$P(X = x) = P(x \leq X \leq x) = F(x) - F(x -)$$

$$\text{Let } F(x) = \int_{-\infty}^x f(t)dt$$

$$f(x) = F'(x) = \frac{d}{dx} \int_{-\infty}^x f(t)dt$$

2.3 Expectation and Variance

Expectations or mean of X is defined by: (μ_X may not be in R_X)

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i) = \int x f(x)dx$$

Properties of Expectation:

1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$ or $\int g(x)f(x)dx$

Variance

Variance of X is defined as σ_X^2

$$\sigma_X^2 = V(X) = E[(X - \mu_X)^2] = E(X^2) - [E(X)]^2$$
$$= \sum_{x \in R_X} (x - \mu_X)^2 f(x) \text{ or } \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$$

Standard deviation of X is defined as σ_X

$$\sigma_X = \sqrt{V(X)}$$

Properties of Variance:

- $V(aX + b) = a^2 V(X)$

3 Joint Distribution

- **n-dimensional random variable, $(X, Y, \dots)$** = $(X(s), Y(s), \dots) \forall s \in S$
- **Range Space** $R_{X,Y} = \{(x, y) \mid x = X(s), y = Y(s), \forall s \in S\}$
- **Discrete / Continuous** two-dimensional random variable depends if **both** random variable is countable and finite or vice-versa.

3.1 Joint, Marginal and Conditional Distribution

Joint probability (mass) function, jpf is defined by $(x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

Well-defined jpf X, Y of both discrete and continuous random variable satisfies:

1. $f_{X,Y}(x, y) \geq 0, \forall (x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0, \forall (x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)dx dy = 1$

Marginal Probability Distribution

Marginal Probability Distribution of X is defined as sum of $f(y)$ after fixing $X = x$

$$f_X(x) = \sum_y f_{X,Y}(x, y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$$

- $f_X(x)$ satisfies all properties of probability function

Conditional probability Function

Conditional Probability function of Y given $X = x$ is defined as

$$f_{Y \mid X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

- $f_{Y \mid X}(y \mid x)$ is defined only for x such that $f_X(x) > 0$
- $f_{Y \mid X}(y \mid x)$ is not a probability function and therefore, sum $\neq 1$

Applications:

- $P(Y \leq y \mid X = x) = \int_{-\infty}^y f_{Y \mid X}(t \mid x)dt$
- Regression function $E(Y \mid X = x) = \int_{-\infty}^{\infty} y f_{Y \mid X}(y \mid x)dy$

3.2 Independent Random Variables

Random variable $X \perp Y$ are **independent** $\Leftrightarrow$

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

$\Leftrightarrow$ both of the following holds

- $R_{X,Y}$ spans **Product Space** $R_{X,Y} = \{(x,y) \mid x \in R_X, y \in R_Y\} = R_X \times R_Y$, when probability function is positive.
- $\forall (x,y) \in R_{X,Y}$

$$f_{X,Y}(x,y) = c \cdot g_1(x) \cdot g_2(y)$$

where g_1 depends only on x , g_2 depends only on y , and c is constant

Let $X \perp Y$

- $P(X \in A; Y \in B) = P(X \in A)P(Y \in B)$
- $P(X \leq x; Y \leq y) = P(X \leq x)P(Y \leq y)$
- Let $g1(\cdot), g2(\cdot)$ be arbitrary functions, $g1(X), g2(Y)$ are independent
- $f_X(x) > 0 \Rightarrow f_Y \mid x(y \mid x) = f_Y(y)$

3.3 Expectation, Covariance and Correlation Coefficient

Expectation, $E(g(X,Y))$:

$$E(g(X,Y)) = \sum_x \sum_y g(x,y)f_{X,Y}(x,y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)f_{X,Y}(x,y)dydx$$

Covariance, $cov(X,Y)$

$$cov(X,Y) = \sum_x \sum_y (x - \mu_x)(y - \mu_y)f_{X,Y}(x,y)$$
$$= E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y$$

Properties of covariance:

- $X \perp Y \Rightarrow cov(X,Y) = 0$, converse is not true
- $X \perp Y \Rightarrow E(XY) = E(X)E(Y)$
- $cov(aX + b, cY + d) = ac \cdot cov(X,Y)$
- $cov(X,Y) = cov(Y,X)$
- $V(aX + bY) = a^2V(X) + b^2V(Y) + 2ab \cdot cov(X,Y)$

Correlation Coefficient, $\rho(X,Y)$

$$\rho(X,Y) = \frac{cov(X,Y)}{\sigma_X\sigma_Y}, -1 \leq \rho(X,Y) \leq 1$$

Properties of joint-variance:

- $V(X + Y) = V(X) + V(Y) + 2 \cdot cov(X,Y)$
- $V(X \pm Y) = V(X) + V(Y)$
- $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n) + 2 \sum_{j>i} cov(X_i, X_j)$
- $V(X_1 \pm X_2) = V(X_1) + V(X_2)$

4 Probability Distribution

4.1 Discrete Probability Distribution

Distribution Name	X	pmf	E(x)	V(x)
Discrete uniform distribution, U(k)	none	$f(x) = \begin{cases} \frac{1}{k} & x \in R_d \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{k} \sum_{i=1}^k x_i$	$(\frac{1}{k}) \sum_{i=1}^k x_i^2 - \mu_X^2$
Bernoulli Trial	$X = \begin{cases} 1 & \text{if bokeh} \\ 0 & \text{if fail} \end{cases}$	$f(x) = p^x(1-p)^{1-x}$	p	$pq = p(1-p)$
Binomial Distribution, Bin(n, p)	X=Number of success	$f(x) = \binom{n}{x} p^x q^{n-x}$	np	npq
Negative Binomial Distribution, NB(k, p)	X=Number of trial	$f(x) = \binom{x-1}{k-1} p^k q^{x-k}$	$\frac{k}{p}$	$\frac{(1-p)k}{p^2}$

Geometric Distribution, Geom(p) = NB(1, p)	X=Number of trial till 1st success	$f(x) = (1-p)^{x-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson Distribution, Poisson(λ)	X=Number of event occurred in fixed time, λ =Expected number of occurrence	$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ
Poisson Process, Poisson(αT)	X=Number of event occurred in fixed time	$f(x) = \frac{e^{-(\alpha T)} (\alpha T)^x}{x!}$	αT	αT

Approximate Binomial using Poisson Distribution

As $n \rightarrow \infty$ and $p \rightarrow 0$ but $\lambda = np$ remains constant, We may approximate $X \sim \text{Bin}(n, p)$ with $X \sim \text{Poisson}(n, p)$

$$\lim_{p \rightarrow 0; n \rightarrow \infty} P(X = x) = \frac{e^{-np}(np)^x}{x!}$$

Condition to use:

$$n \geq 20 \wedge p \leq 0.05 \text{ or } n \geq 100 \wedge np \leq 10$$

4.2 Continuous Probability Distribution

Continuous Uniform Distribution, $X \sim U(a,b)$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}, \quad E(x) = \frac{a+b}{2}, \quad V(X) = \frac{(b-a)^2}{12}$$
$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

Exponential Distribution, $X \sim \text{Exp}(\lambda)$

Model the waiting time until the next event when events occur independently at a constant average rate. For $\lambda > 0$, λ is the average event rate, Mean waiting time is $E(X) = \frac{1}{\lambda}$.

$$f_{X(x)} = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}, \quad E(X) = \frac{1}{\lambda}, \quad V(X) = \frac{1}{\lambda^2}$$
$$F_{X(x)} = P(X \leq x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Hence, for $x \geq 0$,

$$P(X > x) = e^{-\lambda x}$$

Let $\mu = \frac{1}{\lambda}$. Then

$$f_{X(x)} = \begin{cases} \frac{1}{\mu} e^{-\frac{x}{\mu}} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$
$$E(X) = \mu, \quad V(X) = \mu^2, \quad F_{X(x)} = 1 - e^{-\frac{x}{\mu}}$$

Normal Distribution, $X \sim N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, E(X) = \mu, V(X) = \sigma^2$$
$$Z = \frac{X - \mu}{\sigma}$$
$$\varphi(z) = f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$
$$P(x_1 < X < x_2) = \varphi\left(\frac{x_2 - \mu}{\sigma}\right) - \varphi\left(\frac{x_1 - \mu}{\sigma}\right)$$
$$P(Z < z) = P(Z \geq -z) = 1 - \varphi(-z)$$

Quartile α

$$P(Z \geq z_\alpha) = \alpha$$

Approximate Binomial using Normal Distribution

As $n \rightarrow \infty$ and p is constant, we may approximate binomial distribution using normal distribution.

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} = \frac{X - np}{\sqrt{np(1-p)}}$$

Use when $np > 5$ and $n(1-p) > 5$

Continuity Correction

$$P(X = k) \approx P\left(k - \frac{1}{2} < X < k + \frac{1}{2}\right)$$
$$P(a \leq X \leq b) \approx P\left(a - \frac{1}{2} < X < b + \frac{1}{2}\right)$$
$$P(a < X \leq b) \approx P\left(a + \frac{1}{2} < X < b + \frac{1}{2}\right)$$
$$P(a \leq X < b) \approx P\left(a - \frac{1}{2} < X < b - \frac{1}{2}\right)$$
$$P(a < X < b) \approx P\left(a + \frac{1}{2} < X < b - \frac{1}{2}\right)$$
$$P(X \leq c) = P(0 \leq X \leq c) \approx P\left(-\frac{1}{2} < X < c + \frac{1}{2}\right)$$
$$P(X > c) = P(c < X \leq n) \approx P\left(c + \frac{1}{2} < X < n + \frac{1}{2}\right)$$